

Varenya Sri Mudumba

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Education

Texas A&M University, College Station, TX

Aug 2025 - May 2027

M.S. in Computer Science - GPA 4.0/4.0 (Expected May 2027)

National Institute of Technology, Durgapur

Dec 2020 - May 2024

B.Tech in Computer Science and Engineering - GPA 8.84/10.00

Technical Skills

Languages: Python, C/C++, Go, Rust, SQL, Bash.

LLMs & Fine-Tuning: PyTorch, LoRA / DoRA, GRPO / RLVR, RAG, tool-calling / MCP, LangGraph.

ML Techniques: transfer learning, quantization, knowledge distillation, ONNX, EvalPlus.

Training & Infra: HPC / Slurm, multi-GPU / distributed training, AWS, Amazon Bedrock, GCP.

Tooling: Docker, Kubernetes, GitHub Actions, LanceDB, Git, Linux, CI/CD.

Models & Benchmarks: Qwen2.5-Coder, BERT / GPT-2, DistilBERT, HumanEval, MBPP, GLUE.

Experience

Research Intern

Jul 2026 - Aug 2026

Corelight

Remote, USA

- Engineered a self-supervised multi-agent system (Claude on Amazon Bedrock) with tool-calling over a custom 23-tool static-analysis MCP server; agents iterate on their own transcripts to analyze files and auto-generate YARA rules.
- Iteratively refined the agents' prompting and tool orchestration to lift output coverage from 40-60% to 80-98% and cut cost from ~\$10 to ~\$4 per family run, across a ~38K-sample corpus (138 families, 6 file types).

Graduate Researcher

Dec 2025 - Present

Texas A&M University (GAIA Group; Botacin Lab)

College Station, TX

- Researching streaming algorithms and ML methods for malware analysis under Dr. Marcus Botacin's Lab.
- Scaling transfer-learning models over multi-spectral data on HPRC clusters for the GAIA precision-agriculture group.

Software Development Engineer (Intern to SDE)

Feb 2024 - Aug 2025

Zscaler

India

- Built a Pytest platform running 900+ daily regression tests and automated AWS/GCP provisioning (Terraform/Ansible), cutting environment setup from 18 hours to 3 and catching regressions within hours of each commit.
- Refactored legacy cloud modules with Composite and Factory design patterns, cutting technical debt by ~50%.
- Shipped Zero-Trust (ZPA) access-control features adopted by 7+ internal teams and reviewed 70+ pull requests.

Projects

Python Code-Generation SLM

Jun 2026

- SFT:** fine-tuned Qwen2.5-Coder-1.5B with DoRA on a curated corpus (CPython docs, issue-to-fix and CVE-to-patch pairs), reaching **70.7% HumanEval** pass@1, +8.2 over base on held-out spec-to-code.
- GRPO (RLVR) + RAG:** added an execution-grounded reward stage (**MBPP 72.5%**) and a hybrid retrieval layer (Qwen3 embeddings + LanceDB); shipped 2 models and an eval dataset on Hugging Face.

LLM Self-Correction (Supervised Fine-Tuning Study)

Feb - Apr 2026

- Ran LoRA SFT on Qwen2.5-Coder-7B over CodeContests to probe self-correction behavior.
- Benchmarked in- vs out-of-distribution generalization (**HumanEval 57.9%** vs 50.6% base) across MBPP and Codeforces, exposing the brittleness of SFT-based self-correction under distribution shift.

ZTBI - On-Device Prompt-Injection Detector

Nov - Dec 2025

- Built a Chrome extension running **DistilBERT on-device** (ONNX Runtime Web) at **sub-second** latency to flag prompt injection and PII exfiltration in untrusted page content, with no data leaving the browser.
- Engineered a hybrid scoring engine hardened against tag-splitting and asynchronous exfiltration.

Additional

- Publication:** "PSO-Driven Test Pattern Generation for Hardware Trojan Detection," IEEE 2024. DOI.
- Merged 2 PRs into IBM's qiskit-machine-learning; peak CodeChef rating 1893, 500+ problems solved.
- Work Authorization:** F-1 OPT (STEM) - up to 36 months, no immediate sponsorship.